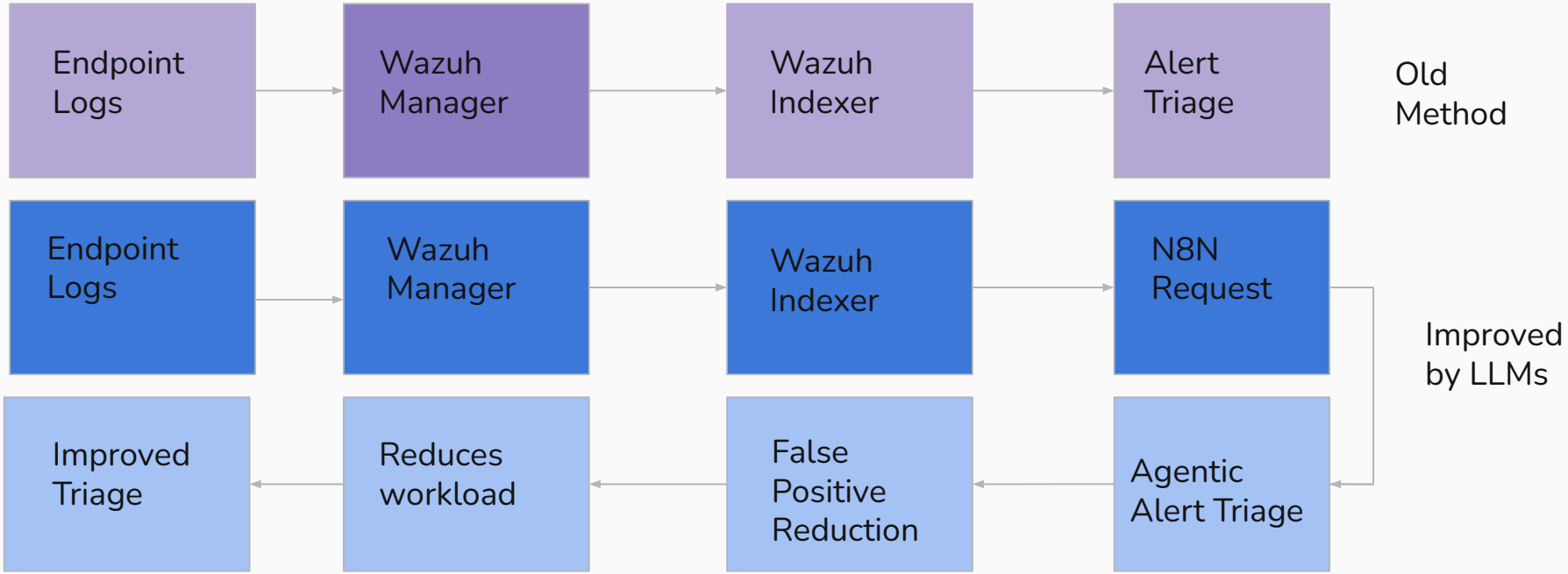
SOC System Architecture



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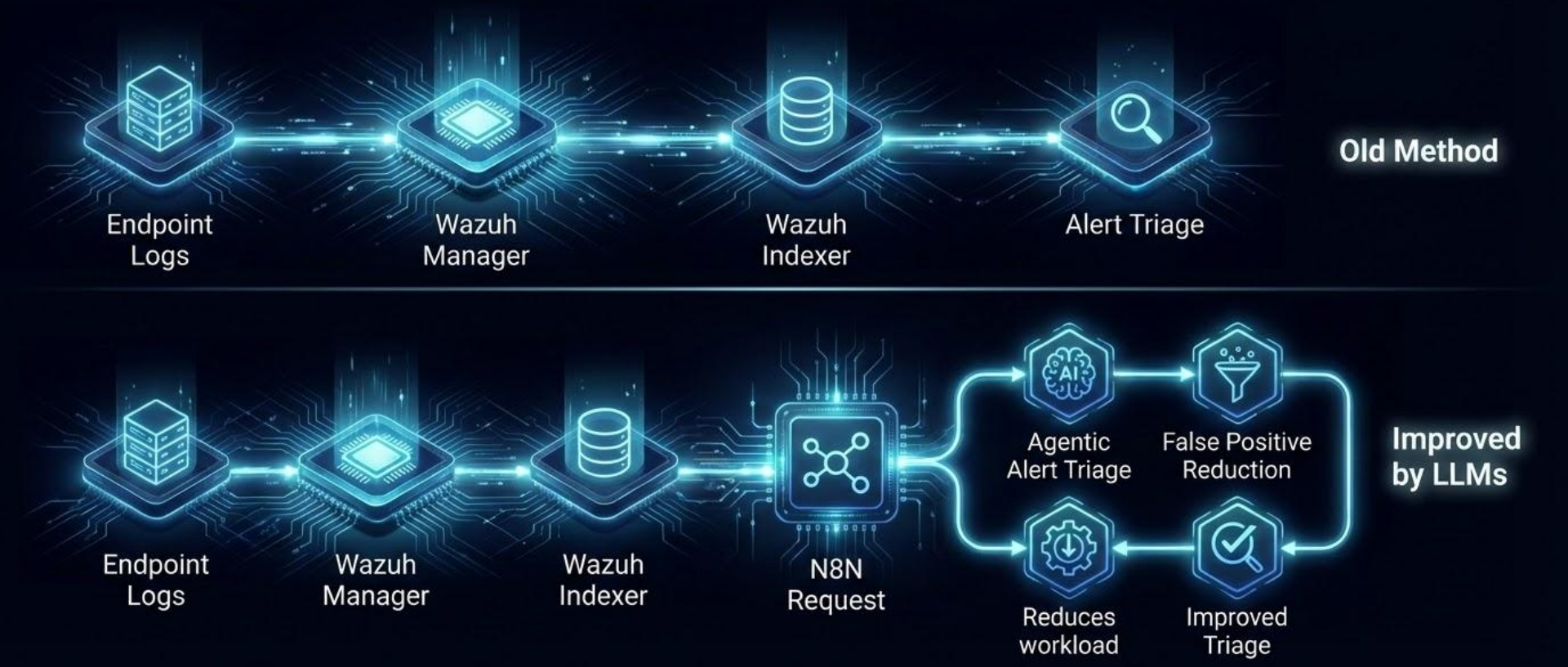


Current Stage of Data

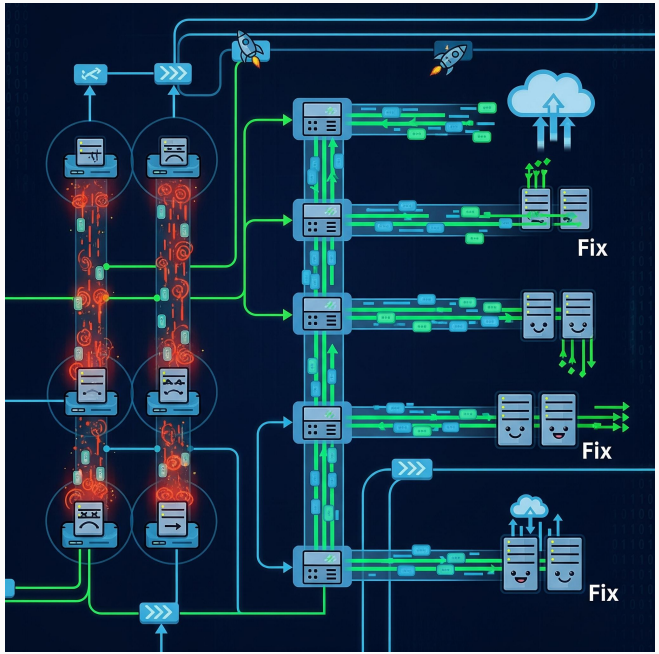
- MITRE ATT&CK STIX Database and TAXII server logs.
- Digital Forensics Artifact Knowledge Base - Digital Forensics Artifact Knowledge Base
- Windows Events Attack Samples - Windows Events Attack Samples
- Windows Registry Knowledge Base - Windows Registry Knowledge Base
- Publicly available PCAP files
- Malware Traffic Indicators
- DEFEND RESOURCES

Current Stage of Tools

- Incident Response toolkit for simulation and detection of threat actors.
- Environment for the simulation of attacks and advance logging mechanism using auditd and wazuh's cluster of SIEM tools.
- n8n workflow for the HTTP request from the LLM to wazuh and alert triage using the fine tuned LLM to reduce the false positives and improve the alert triages.
- [Incident response and triage tools and softwares.]
- Active response from virustotal API key for known hashes and malicious files.

System Bottlenecks and Fixes

- Systems are not capable of running the Attacker, defender and server(log collector) virtual machines at one time wazuh consumes 8GB of RAM single handedly.
- RAM consumption is 100% (16 GB) in running the VMs before running the attacks.
- Tried on Google Collab and Jupyter notebook but model is less complex and due to less VRAM there are hallucinations in the output.
- Without proper GPU the LLM is almost impossible to train.



GPU Requirements Justified

Things that are Completed	Things that are needed to be completed.

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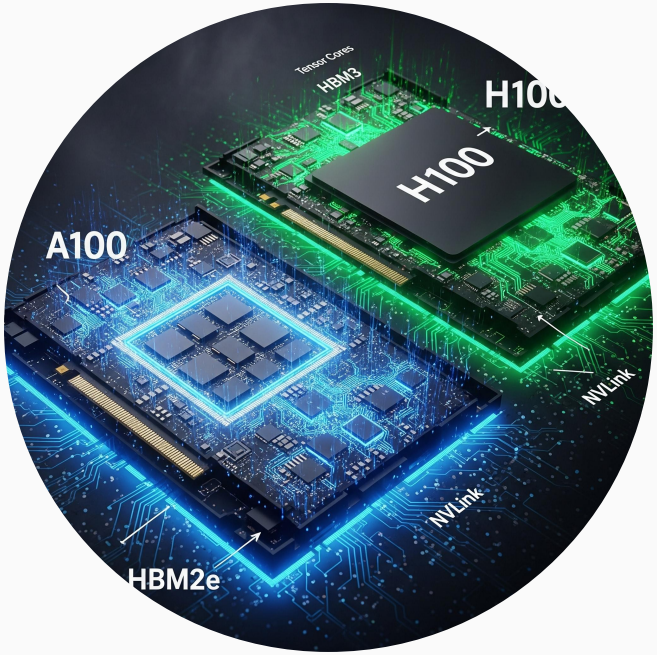
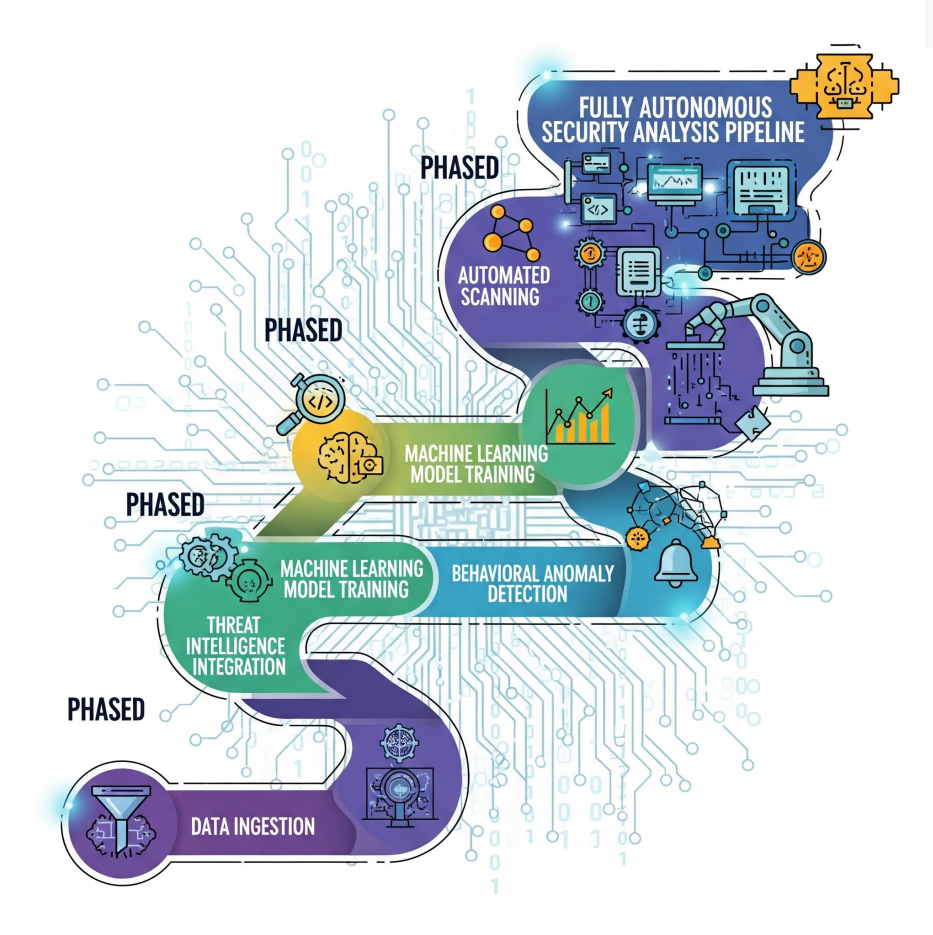


- Finetuning LLMs needs deep GPU memory for context
- High-dimensional embeddings require significant VRAM capacity
- Sequence lengths exceeding 4k tokens demand powerful GPUs
- A100 or H100 are ideal for continuous online retraining
- RTX 4090 offers excellent performance at a lower cost

Alternative Compute Options

- Utilizing vLLM or 4-bit models on consumer cards
- Cloud GPU rentals offer flexible, temporary compute resources
- AWS Bedrock provides accessible, powerful managed AI services
- Less accuracy and high latency are key performance tradeoffs
- Sensitive logs face data residency and privacy risks in the cloud

Implementation Plan Roadmap

- Phase 1 focuses on initial deployment and testing
- Next phase implements AI components and active response
- Future work includes full SOC integration and scaling efforts
- Final goal is a fully autonomous security analysis pipeline